

Navigating Daily Life: Insights from Powered Wheelchair Users on Assistive Technologies and Caregiver Support

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ABSTRACT

The increasing number of powered wheelchair users, driven by an aging global population and advancements in medical care, highlights a critical need to understand their unique experiences and challenges, particularly regarding assistive technologies and caregiver support. Our study, drawing on qualitative data from 11 participants with diverse disabilities, investigates the complex interplay of technology, independence, and socio-economic factors in their lives. It reveals a significant gap between the potential and actual use of assistive technologies, underscoring the importance of innovation in areas where users heavily rely on caregivers. The implication of this study includes emphasizing the necessity for inclusive, adaptable, and economically accessible assistive technologies and highlighting the vital role of human support in augmenting the quality of life for individuals with mobility impairments.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility**.

KEYWORDS

Powered Wheelchair User Experience, Caregiver Burden, Assistive Device, Disability Support

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1 INTRODUCTION

The landscape of disability and mobility impairment is experiencing a profound transformation, reflective of broader demographic and medical trends. In the United States, over 61 million adults live with some form of disability, with a significant proportion (13.7%) experiencing mobility challenges that impact their independence and daily living [2]. Specifically, mobility impairments, often resulting from conditions such as central nervous system disorders or stroke, affect millions, with a majority requiring the use of assistive devices, including over 1.7 million wheelchair users. This growing reliance on mobility aids, coupled with difficulties in performing activities of daily living (ADLs) independently, underscores a critical concern. Around one-third of mobility device users require assistance for one or more ADLs [30]. Moreover, the number of wheelchair users has grown annually by 5.9%, highlighting an increasing need for adequate support [1]. Currently, these supports include a combination of caregiver assistance and assistive devices, but independence for those with limb dysfunctions stands crucial. So, exploring and understanding the experiences of powered wheelchair users is vital for enhancing their quality of life.

In recent research in Human-Computer Interaction (HCI), significant strides have been made in understanding the technological aspects and interaction preferences of wheelchair users, such as the development of wheelchair-based computing platforms [16, 17, 27, 48], and the choice of computing devices by power wheelchair users [17], there remains a lack of comprehensive understanding of the daily realities and diverse needs of this population. For instance, studies have explored gesture preferences for interactive systems [10] and accessible interactions with public displays [46], yet the broader context of users' everyday lives and socio-economic challenges have been less examined. In this research, we aim to gain a thorough understanding of powered wheelchair users' experience, perspectives, and challenges of their naturalistic use of assistive technologies (ATs) and outline the challenges they face while using them in various stages of their lives by answering the following research questions:

- **RQ1:** How do powered wheelchair users navigate the balance between independence and assistance in their daily lives?

- **RQ2:** What are the specific challenges and adaptations associated with the use of ATs by powered wheelchair users?

We approached these questions by conducting unstructured interviews with eleven ($n=11$) powered wheelchair users with varied disabilities, including spinal cord injury, cerebral palsy, muscular dystrophy, and other physical disabilities. We also dig into the complex balance between users' desire for independence and their reliance on assistance, revealing a broad spectrum of self-reliance and the vital role of caregivers. Our analysis identifies technological gaps, with an apparent demand for more advanced, versatile, and user-friendly technologies. We also explore the cognitive and emotional aspects of users' strategic planning for daily activities, highlighting the intricate relationship between technology, personal tools, and social dynamics. It underscores the necessity for developing inclusive, adaptable, and economically accessible assistive technologies and advocates for the crucial role of human support in improving the quality of life for individuals with mobility impairments, thereby enriching the discourse in the HCI community.

2 RELATED WORKS

Disability studies and assistive technology significantly influence powered wheelchair users, necessitating a detailed study of these technologies and caregiver support. Powered wheelchairs impact more than mobility, intertwining with users' lives and challenges, highlighting the need for thorough research considering caregivers' roles [5, 23]. The objective here is to scrutinize the progress in technology, policy, and their impacts, aiming to fill crucial knowledge gaps in the field.

2.1 Historical Evolution and Persisting Challenges for Powered Wheelchair Users

Assistive technologies for powered wheelchairs have evolved from basic early models to advanced, smart-tech devices, enhancing user autonomy and safety. This evolution, starting post-World War II, has transitioned from simple motorized units to modern wheelchairs with AI, machine learning, voice control, GPS, and smart connectivity [24][34][26]. However, the use of powered wheelchairs faces various challenges, including socio-economic disparities [21][19], limited access in rural areas [37], urban infrastructure not accommodating wheelchairs [48][39], and psychological factors like social stigma [33][22]. These issues highlight the need for a holistic approach in wheelchair design and support, emphasizing social inclusion and mental well-being.

Powered wheelchair users face challenges due to mismatches between wheelchair design and urban infrastructure, impacting access and autonomy [7][20]. While technological advancements have improved autonomy, they also bring complexities in maintenance [40][45]. Psychosocial factors like stigma and isolation, alongside economic disparities, further influence their experiences, underscoring the need for inclusive design and support systems [31][12].

Caregivers, integral to the operation and emotional support of powered wheelchair users, play roles as facilitators and intermediaries in assistive technology use [25]. Their proficiency and attitudes greatly affect the user's autonomy and experience [44].

However, they face challenges like physical strain, emotional stress, and financial burdens, compounded by the need for continuous education on evolving technologies [28][47].

2.2 Underexplored Areas, Theoretical Frameworks, and Methodological Reflections

Current research on powered wheelchair users often overlooks their holistic experiences, particularly in psychosocial well-being and adaptation, focusing mainly on technological aspects [39]. There's a lack of studies on the emotional and social impacts of wheelchair use, including mental health and social integration, as well as on the experiences of caregivers [8][36]. Additionally, socio-economic factors including economic status, insurance coverage, geographical location, and environmental accessibility are insufficiently explored [15][42][35]. Research on powered wheelchair users employs various frameworks: the Social Model of Disability [41] emphasizes societal barriers over individual impairments; Technology Acceptance Models [6] focus on users' acceptance of technology based on perceived usefulness; the Caregiver Stress Model [32] examines the stressors of caregiving across different contexts; Bronfenbrenner's Ecological Model [18] analyzes environmental influences on caregiving; and the Intersectionality Framework considers the compounded challenges due to intersecting identities. These collectively inform policies and interventions for better accessibility and quality of life for those with disabilities. Historically, research on powered wheelchair users has predominantly used quantitative methods for demographic analysis [13], often overlooking individual experiences. In contrast, qualitative methods offer depth but lack wider applicability [29][42].

The present study, thus, focuses on qualitative analysis of the powered wheelchair users' daily lives around ATs, aiming to gain a comprehensive understanding of their experiences, challenges, and preferences in navigating daily activities and interactions within their circumstances.

3 METHODS

We interviewed 11 diverse powered wheelchair users through unstructured interviews, employing a conversational approach to thoroughly investigate their experiences, specifically focusing on their interactions with ATs and caregiver support. The study took place in the Midwestern region of the United States. The choice of unstructured conversational interviews was informed by their sensitivity to users' privacy and their proven reliability over structured interviews in different studies [9, 11], ensuring a respectful and effective data collection process.

3.1 Participants and Study Procedure

Our data collection employed an unstructured interview format, facilitating a conversational approach to gather deep insights from participants. These conversations were conducted as part of participant recruitment and screening for a funded project focused on developing an advanced assistive robot designed for activities of daily living. We avoided direct questioning, inviting participants to openly share their experiences and any pertinent aspects of their

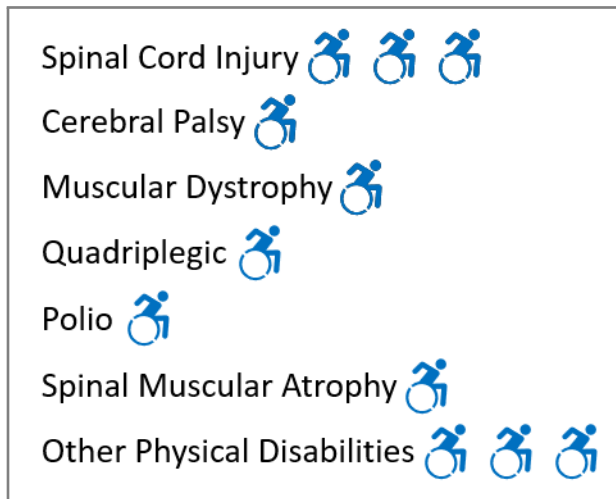


Figure 1: Participants with diverse physical disabilities.

daily lives. This method allowed for authentic, self-expressed narratives, providing a comprehensive view of their challenges, use of assistive technology, and support systems.

Our study engaged 11 participants, each bringing a unique perspective from a spectrum of disabilities, including Spinal Cord Injury (n=3), Cerebral Palsy (n=1), Muscular Dystrophy (n=1), Quadriplegic (n=1), Polio (n=1), Spinal Muscular Atrophy (n=1), and other physical disabilities (n=3), ensuring a broad representation of experiences. This diversity ensures a wide-ranging representation of experiences within our research. Our selection of a small, diverse sample was a deliberate strategy designed to facilitate a deep dive into each participant's unique experiences. This method aligns with the principles of qualitative research, which value the depth and richness of data over sample size. This approach ensured that the data authentically represented the participants' real-world experiences and laid a robust foundation for our analysis.

3.2 Data analysis

The qualitative data analysis from unstructured interviews with powered wheelchair users was executed precisely to uncover deep insights into their daily experiences. Employing thematic coding [14] suited for the conversational nature of our data collection, we delved into the narratives to identify and interpret patterns and themes. The initial phase involved an immersive reading of all interview transcripts to grasp the content fully. This process led to identifying preliminary themes that naturally surfaced from the participants' stories. We engaged in systematic coding, tagging key phrases, anecdotes, and statements, which laid the groundwork for a refined thematic framework. This step was iterative, constantly comparing and emerging themes to accurately reflect the participants' experiences and viewpoints. The resultant qualitative data analysis transcended mere categorization of information; it delved into the depth and complexity of the participants' lives. The synthesized themes provided a comprehensive narrative of the daily realities of powered wheelchair users, forming the cornerstone of our study's findings. Our systematic approach revealed

key themes meticulously organized into a thematic framework, comprising 'The Dynamics of Wheelchair-Dependent Living' and 'Technological Interface and User Experience.' These themes, further detailed in Section 4, serve as the foundation for understanding the multifaceted experiences of powered wheelchair users, from daily challenges and social interactions to the impact of assistive technologies on their lives.

3.3 Ethical Considerations and Positionality

Due to the sensitive nature of the data, we diligently addressed the ethical implications of our research to safeguard the rights of the participants. The conversations, integral to our data collection, were conducted as part of participant searching, recruitment, and screening for a funded project. University of Wisconsin – Milwaukee's Institutional Review Board (IRB) assessed and authorized all studies under this project. We took special care to ensure that participants were comfortable during the conversation and were only obliged to share information within their comfort level.

The research team's positionality was a crucial aspect of this study. Our team comprised members with direct connections to the world of disability. One member had extensive experience as a caregiver for a family member with a disability, while others are experts in assistive and rehabilitation technologies. This diversity in backgrounds enriched our understanding and sensitivity toward the subject, ensuring a more empathetic and informed approach to the study.

4 FINDINGS

The findings of our study, derived from a meticulous qualitative analysis of the data, present a comprehensive understanding of the lived experiences of powered wheelchair users. The analysis revealed intricate details about their daily lives, highlighting both the challenges and adaptations they encounter. These findings paint a vivid picture of the realities faced by individuals who rely on wheelchairs for mobility, shedding light on the complexities of wheelchair-dependent living. The insights gathered provide a nuanced perspective on how assistive technologies intersect with users' experiences, capturing the essence of their interactions with technology and the impact on their quality of life.

4.1 The Dynamics of Wheelchair-Dependent Living

4.1.1 Everyday Challenges and Adaptations. Our qualitative analysis illuminated a range of everyday difficulties, from physical barriers in the environment to limitations imposed by wheelchairs. Participants shared experiences navigating spaces not designed for wheelchair access, highlighting the constant need for attention and assistance. Additionally, the interviews revealed how wheelchair users often have to modify or personally customize their wheelchairs, underscoring the lack of one-size-fits-all solutions in current assistive technologies. A recurring theme was the challenge of performing basic activities of daily living (ADLs) that able-bodied individuals might take for granted. This included tasks like reaching for objects, managing personal care, or engaging in social activities. However, alongside these challenges, participants shared various adaptive strategies they developed. These ranged

from employing specific techniques to maneuver their wheelchairs in tight spaces to using technology and relying on caregiver support for certain tasks. For example, participant 2 (P2) mentioned that:

"I must be strategic about the accommodation I might need each day before coming to the office..."

Despite the array of challenges, their ability to adapt and find personalized solutions speaks to their determination to maintain autonomy and engage actively in their lives. These insights are crucial in understanding the everyday realities of wheelchair-dependent living, providing a foundation for enhancing the design and functionality of assistive technologies.

4.1.2 Independence and Assistance. In our study, the contrast between the desire for independence and the need for assistance was a prominent theme among powered wheelchair users. Participants expressed a strong inclination towards autonomy, highlighting the significance of performing daily tasks and making independent decisions. However, the reality often involved reliance on support from human caregivers. The extent of this dependence varied, with the average caregiver assistance needed being approximately 6.8 hours daily, reaching up to 12 hours for some, while one participant required no assistance at all. This variability in independence levels among the participants was primarily influenced by the nature of their disability, the capabilities of their wheelchairs, and the accessibility of support systems, with two participants reporting high independence due to advanced wheelchair features or other assistive devices. P8 said that:

"The power chair is very important. I would either have to have a massive amount of help or not be able to live independently if I didn't have the capacity to use the wheelchair like the one I do..."

In contrast, others (n=6) faced significant limitations and depended heavily on caregivers for basic needs, highlighting the disparities in independence within this community. This paradox underscores the importance of designing assistive technologies that are functionally effective and cater to the diverse needs and preferences of wheelchair users. It also points to the need for comprehensive support systems that respect and enhance the autonomy of individuals while providing the necessary assistance to enable a fulfilling and active life. Understanding this interplay between independence and assistance is crucial for developing solutions that empower powered wheelchair users.

4.1.3 Everyday Life and Social Interactions. In exploring the everyday lives of powered wheelchair users, our study illuminated how their assistive devices deeply influence their social interactions and community engagement. The participants provided valuable insights into the dual role of wheelchairs in facilitating and sometimes hindering social activities. One notable aspect of these interactions is the impact of technology on social inclusivity. Current assistive technologies, such as the iBot wheelchair [3], have made strides in enhancing social engagement for wheelchair users. A feature appreciated by one of our participants is the Eye Level View offered by such devices. This functionality addresses a crucial aspect of social interaction – engaging with others at eye level. P10 poignantly highlighted this:

"Having the eye level view with others during a conversation is very important for me, and my iBot can do that..."

This ability to interact face-to-face, rather than from a lower vantage point, significantly enhances the quality of social interactions and contributes to a sense of inclusivity. However, our participants' experiences also revealed the complexities of social life as a powered wheelchair user. While some reported positive experiences where their mobility devices facilitated social interactions, others faced challenges in accessing public events, leading to feelings of isolation. The experiences shared by participants also underscored the significance of social support systems, including family, friends, and caregivers, in enhancing their quality of life. These support systems often played a crucial role in facilitating social engagement, underscoring the importance of an inclusive community approach to disability. Understanding these dynamics is crucial for developing policies and designing spaces that are truly inclusive and accommodating of all individuals.

4.1.4 Financial Burdens and Insurance Complexities. The exploration of the financial implications and insurance complexities faced by powered wheelchair users in our study revealed a significant economic burden that profoundly affects their access to necessary assistive technologies. This burden was exemplified in the costs associated with powered wheelchairs and the high expenses of additional assistive devices like the iBot and the robotic assistive arm, Jaco [4]. Jaco is an advanced robotic arm designed to assist wheelchair users in performing various tasks. However, many participants expressed that despite the potential life-enhancing benefits of such technology, it remains inaccessible due to its prohibitively high cost. This situation is exacerbated by the inconsistent coverage provided by insurance policies, which often do not extend to cover these advanced assistive technologies. P9 mentioned:

"I tried Jaco once. It just wouldn't help me enough for the really incredibly high price..."

The disparity in insurance coverage and the resultant financial strain were consistent themes among our participants. While some had coverage for their wheelchair costs, assistive devices, or caregiver costs, others were left part of the cost to shoulder them, highlighting significant gaps in healthcare support systems. This financial challenge extends beyond the technology acquisition and encompasses the ongoing maintenance and upgrades necessary for these devices. These findings underscore the urgent need for more inclusive and equitable healthcare policies that provide comprehensive coverage for essential assistive technologies. Addressing these economic and insurance-related challenges is critical in ensuring equitable access to technologies like the iBot and Jaco, which have the potential to dramatically enhance the independence and quality of life for powered wheelchair users.

4.2 Technological Interface and User Experience

4.2.1 User Perspectives on Current Technologies. Our study on user perspectives regarding current assistive technologies revealed that their usage among powered wheelchair users is minimal. Of the 11 participants, only three utilized advanced technologies such as

The Role and Impact of Assistive Technologies	User-Centric Challenges and Adaptations
<i>P6: I guess, and I just look at Jaco, it's just about an extension of me...</i>	<i>P8: Maintenance is a problem because the wheelchair repair people usually will not come to the home...</i>
<i>P1: I find that it makes life a whole bunch easier. I just, it gives me a sense of independence...</i>	<i>P9: Toileting is, I think, my personal number one barrier to independence... Transportation is my second biggest barrier to full independence...</i>
<i>P6: I hate to get out of my chair because my Jaco goes on my chair. I wish I could pick it up and put it on my bed... I feel very vulnerable at night. Because I don't have my hand anymore...</i>	<i>P6: So I've had to improvise. Like I spent a lot of time on Amazon finding my own plates and cups to practice with that...</i>

Table 1: Notable quotes from the participants on the role and impact of assistive technologies and user-centric challenges.

robotic arms like Jaco or the iBot. The majority, including six participants, were either unaware of or had never used these technologies, relying on their wheelchair's functions and caregiver support. This limited use and awareness of advanced assistive devices indicate a significant disconnect between technological advancements and their adoption by wheelchair users, often constrained by limited information and accessibility. This scenario underscores a crucial gap between the availability of assistive technologies and the actual needs and awareness of the users.

4.2.2 The Role and Impact of Assistive Technologies. The insights gained from our participants highlighted the profound role and impact of assistive technologies in the lives of powered wheelchair users. These technologies, ranging from advanced wheelchair models like the iBot to robotic arms with chin, finger joysticks [38], or eye-gaze control [43], are not just tools for mobility but are integral to fostering independence and enhancing the quality of life. Participants described these technologies as integral to their identity and daily living, with one expressing the technology's role in independence and another describing it as an extension of their body, as described in Table 1.

The significance of these technologies extends beyond daytime activities, as one participant expressed a sense of vulnerability without their device at night [Table 1]. These statements underscore the deep reliance on assistive technologies for physical support and emotional and psychological well-being. These experiences highlight a crucial aspect of assistive technologies: their potential to empower users with a sense of autonomy and security. However, many users' limited access and awareness also indicate a need for broader outreach and availability of these technologies.

4.2.3 User-Centric Challenges and Adaptations. In examining the user-centric challenges faced by powered wheelchair users, our study revealed a complex landscape of adaptation and individualized barriers. These challenges highlight the nuanced interplay between users and their assistive devices, emphasizing the need for

adaptable and personalized technological solutions. One notable challenge encountered by participants relates to the maintenance of their assistive devices. As one user pointedly expressed in Table 1, this underscores a practical issue that extends beyond the technology itself to the services that support it, impacting the users' ability to rely fully on these devices for their daily needs.

Separately, the study illuminated specific ADLs where current assistive technologies fall short. Users identified key areas where dependence on caregivers remains indispensable, as no available technology offers a complete solution. This candid admission highlights a significant gap in assistive technology, where current innovations have yet to adequately address fundamental personal needs. These insights point to the need for targeted technological advancements addressing these specific yet critical aspects of independent living. Similarly, the users highlighted the process of adapting to assistive devices and the need to modify or find workarounds for their equipment personally. Participants illustrated this by sharing their experience with the Jaco robotic arm [Table 1]. This example reflects the users' resilience and resourcefulness and underscores the importance of designing more intuitive assistive technologies customized to individual needs. Again, there needs to be more support and dedication to hook them up with state-of-the-art technologies. P7 reflected on the challenges of integrating technology into their routine, stating,

"There's technology out there, but there is nobody that knows how to hook it up and make it work properly..."

5 DISCUSSION

5.1 Navigating Independence and Assistance in Powered Wheelchair Use

Our findings offer a detailed understanding of how powered wheelchair users navigate the balance between independence and assistance, directly addressing our research question. The reliance on caregiver support, often preferred over advanced assistive technologies, is attributed to a reluctance to adopt new technologies, a lack of support for integrating current technologies, and the high cost of these devices relative to the benefits they provide. This is evident as only a few participants (3 out of 11) were aware of advanced devices like Jaco and iBot, indicating a significant gap in awareness and inclination towards traditional, reliable support methods. Our study finds resonance with existing research, revealing that powered wheelchair users often grapple with socio-economic disparities and caregiver dependencies, which significantly impact their daily lives [21][19]. It also resonates with the Technology Acceptance Model [6], which helps explain the reluctance towards adopting advanced technologies seen in our participants, underlining the importance of designing assistive solutions that are technically advanced, user-friendly, and directly relevant to the users' everyday needs. Psychological challenges like social stigma further complicate their experiences, underscoring the urgent need for more inclusive and empathetic support systems [22, 33].

The study also sheds light on the challenges faced in device maintenance and support services, which impact users' confidence and independence, further exploring the interplay between technological reliance and human assistance. Our findings reveal that current technologies fail to adequately address essential ADLs,

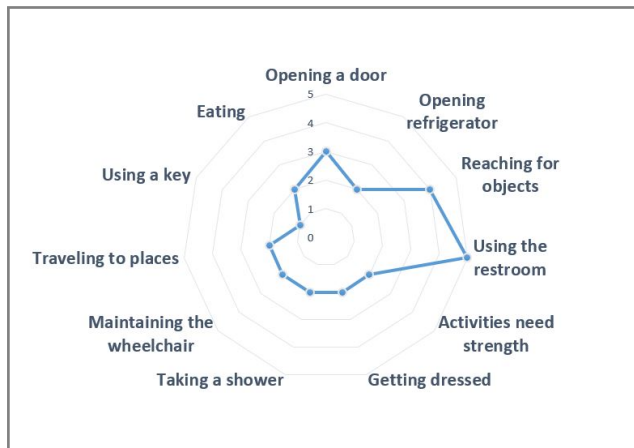


Figure 2: Most challenging ADLs.

such as using the restroom, leading to a continued dependence on caregivers. The list of ADLs that are most challenging to the participants is illustrated in Fig. 2. These challenges align with our second research question, highlighting the need for more intuitive, customizable, and user-friendly assistive technologies. A holistic approach is necessary to meet users' diverse needs and preferences effectively. This approach should consider technological advancement and accessibility, affordability, and suitability for users' real-world needs. By implementing a strategy encompassing user-informed design, improved support services, and the development of inclusive environments, we can facilitate greater empowerment and independence for powered wheelchair users.

5.2 Implications for Future Design and Research on Assistive Technologies

5.2.1 Towards Inclusivity. Future designs should prioritize inclusivity, customization, and user-friendliness, departing from one-size-fits-all solutions to embrace tailored approaches informed by researching user feedback. This user-centric perspective, coupled with a holistic approach, recognizes that challenges faced by users extend beyond technology to contain support services and societal environments. This could involve streamlining maintenance processes, ensuring timely and user-friendly support, and providing accessible resources for ongoing assistance.

5.2.2 Towards User-Driven Development. To move towards a user-driven development approach, it is essential to engage disabled stakeholders actively in the entire development process. This goes beyond mere participation in user surveys; it involves incorporating individuals from the target population as project members. Combining their perspectives, experiences, and insights through research ensures that assistive technologies are finely tuned to meet the distinctive needs, preferences, and real-world challenges of individuals with disabilities, fostering a more inclusive and practical design.

5.2.3 Towards Technological Awareness. Addressing the recognized need for increased interest in advanced assistive technologies requires forthcoming initiatives centered on education, outreach programs, and promotional campaigns to inform users about the advantages and functionalities of advanced assistive technologies. Furthermore, open communication among all the stakeholders is essential to ensure that identified advantages and disadvantages are effectively conveyed to each stakeholder based on their communication preferences, promoting a shared understanding and facilitating informed decision-making processes. Thus, it could ensure a more informed and receptive user base, enhancing the quality of life for powered wheelchair users.

6 CONCLUSION AND FUTURE WORK

This study has offered valuable insights into the experiences of powered wheelchair users, highlighting the vital role of assistive technologies in enhancing independence and underscoring the challenges they face daily. The results of our qualitative research highlight the need for greater awareness and user-focused design, as they show a considerable disconnect between the promise of assistive technologies and their actual use. The results also highlight how crucial assistive technologies are to advance, mainly when users still rely significantly on caregivers. However, it acknowledges limitations in data scope and participant diversity, which might impact the generalizability of the findings. Future research will include broadening its participant base to capture a broader range of experiences, enhancing our understanding of the diverse needs and challenges in different contexts.

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